

Improving Open Source Software Patch Contribution Process: Methods and Tools

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ABSTRACT

The patch contribution process (PCP) is very important to the sustainability of OSS projects. Nevertheless, there are several issues on patch contribution in mature OSS projects, which include time consuming process, lost and ignored patches, slow review process. These issues are recognized by researchers and OSS projects, but have not been addressed. In this dissertation, I apply Kanban method to guide process improvement and tools development to reduce PCP cycle time.

Categories and Subject Descriptors

D.2.8 [Software Engineering]: Metrics – *process metrics*; D.2.9 [Software Engineering]: Management – *life cycle, productivity, software process models*.

General Terms

Management, Measurement

Keywords

Lean methods, kanban, free/open source software, coordination, productivity

1. INTRODUCTION

Open Source Software (OSS) projects and systems have become significant parts of the software economy. The sustainability of an OSS project depends largely on community contributions; especially patch (source code and document change) contributions. The patch contribution process (PCP) is very important to OSS projects because it is the primary quality assurance mechanism, enables learning and knowledge transfer in software projects, and provides an opportunity for recruiting developers. Nevertheless, there are several barriers to contribution to mature OSS projects. Patch contribution is time consuming and slow. Patches are often lost, ignored, under-reviewed, or avoidably rejected. Moreover, for contributors participating in multiple projects differences in processes and tools can create confusion and increase learning time. A systematic understanding

of the PCP is required in order to determine key problems and improve the process.

The goal this dissertation is reducing cycle time of OSS patch contribution by eliminating waste through process improvement and tools. In this research, I define PCP cycle time as the average length of time it will take to resolve a patch. The outcome of this research will be a descriptive model and a set of tools and methods for analyzing and improving the PCP. This dissertation aims to answer three research questions: 1) What are the current practices of OSS PCPs? 2) What are major factors that impact PCP cycle time? 3) How can we reduce PCP cycle time?

2. RESEARCH PROBLEMS

Despite the long history of open source the PCP remains somewhat immature and, has been given little research attention. Using the data collected from ten OSS projects [6], I have identified two common issues that can be attributed by tools, which are described below.

2.1 PCP is Time-Consuming and Slow

The PCP consists of several activities: creation, publication, discovery, review and application. Many of these activities involve several people including patch contributors, reviewers, peer developers and committers, who collaborated asynchronously [6]. Any process of many steps that is executed serially and cooperatively is likely to be difficult and take a long time; the PCP is no exception. As a result, the PCP can be time-consuming and slow.

The amount of time that it takes to complete the process depends on many factors such as contributor experience, the capability of tools and the quality of patches. Processing a patch inevitably takes hours and often takes weeks [4],[7]. Although patch review is an important defect detection mechanism, in some projects only patches created by new contributors are required to be reviewed prior to being committed; a policy intended to reduce the delay and complexity for experienced contributors [4]. Literature has identified the slow nature of the PCP as an issue, but studies have yet to address how current patch review tools address this issue [4].

A better patch review tool can provide information needed when reviewers review patches, thus reducing the time required to complete the task.

2.2 Patches are Often Ignored, and Under-Reviewed

Although OSS projects encourage patch contribution, they have problems handling the amount of patches they receive [4]. In many projects, a significant number of patches did not receive responses [1]. Patches might be ignored unintentionally when the reviewers are too busy with other OSS development tasks. Sometimes they can be ignored because of the poor quality of the submitted patches [2]. Moreover, in many OSS projects, only a small group of people participate in patch review [2],[4]. The average number of reviewer responses per patch posting is less than two and the majority of the patches are reviewed by one person [4],[1].

None of the tools we reviewed provides an effective way for reviewers to discover patches that were not assigned to them. Consequently, the majority of patches are ignored or under-reviewed. Better support for patch discovery and awareness is needed to solve these problems. Making patches easily discoverable should increase the number of peer developers reviewing patches, which will reduce the number of ignored patches and increase the number of reviewers per patch.

3. HYPOTHESES

In this dissertation, I will focus on reducing the cycle time of the PCP. By reducing the cycle time, the PCP will become less time consuming. Therefore, fewer patches will be ignored and more developers will review patches. The following hypotheses will be tested in order to answer the research questions.

Hypothesis 1: PCP cycle time can be reduced by limiting number of work-in-progress at the bottleneck steps.

The cycle time depends on the capacity of bottleneck. By improving the bottleneck, the overall cycle time of the PCP will be decreased. I will use critical chain method [3] to guide the identification of bottleneck and develop tools to improve it.

Hypothesis 2: The amount of time that it takes to review a patch can be reduced when it is performed within the context of the software development task.

The cost of switching from the developers' coding task into doing code review can be so high that busy developers do not participate in code review. The task switching costs can be eliminated by providing the right patch to be reviewed that relates to the developer's task. There are two main benefits from reducing the amount of time it takes to review a patch. The OSS PCP cycle time can significantly be reduced and the number of peer developers who perform can be increased.

4. METHODS AND APPROACH

This dissertation is divided in to two phases. The first phase focuses on using the Kanban metrics to assess the current cycle time of three OSS projects. I will identify the key underlying problems in the PCP. The results of the first phase will be the value stream maps of three OSS projects and a system for collecting metrics for the PCP, a comparative case analysis between the performances of these three projects, and a set of common underlying problems that affect the PCP cycle time. The second phase focuses on improving the PCP of the selected OSS projects by suggesting process change and improved tools. The results from the first phase will be used to develop tools and

suggested process changes that will solve the key problems, hence improving the PCP. I will develop tools from the suggested improvement in the previous phase that will reduce the PCP cycle time. The tools will be developed as enhancements to existing tools used by the selected OSS projects such as IDEs, and patch tracking systems. I will demonstrate that by fixing the right problems, small improvements to the existing process and tools can significantly reduce the PCP cycle time. To increase the possibility that the tools will be used by the projects, the tool improvements will be developed in such a way that it can be deployed into the existing tools used by the selected project. However, if this is not possible, I will conduct evaluations in an experimental classroom setting to demonstrate that the improved tools significantly reduce the PCP cycle time by reducing the amount of time that it takes to discover and review patches.

5. CURRENT STATUS

I have conducted a preliminary study and developed a generic OSS patch contribution process model. The model consists of five main activities: patch creation, publication, discovery, review, and, application. These activities are common across ten OSS projects [6]. I published a list of key enhancements to existing tools that may reduce cycle time of PCP [5]. The goals of the tool enhancements are to shorten the patch review interval and increase number of reviews by making patch easily accessible within the development environment.

6. REFERENCES

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